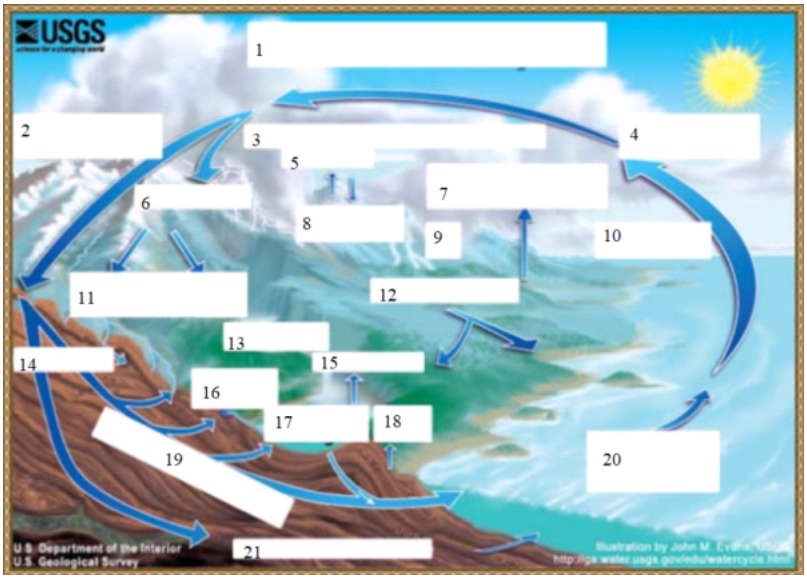
Dynamic Planet C - Dynamic Planet C - Rickards Invitational Div. C - 11-07-2021

- 50 minutes are allowed. Stop precisely when the timer goes off.
- You are allowed to leave the page only to access online notes
- Wrong answers are not penalized.
- Tiebreakers will be marked with an asterisk(*)

Please reach out to the event supervisors through the chat if you have any questions.

Good luck and have fun!



Label the boxes above, the number for the box corresponds to the question. (Hint Number 1 is the water cycle)

1. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Water Cycle

2. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Water storage in ice and sn

3. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Water storage in the atmos

4. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Condensation

5. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Sublimation

6. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Precipitation

7. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Evapotranspiration

8. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Desublimation

9. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Fog Drip

10. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Evaporation

11. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Snowmelt runoff to streams

12. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Surface runoff

13. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Streamflow

14. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Infiltration

15. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Evaporation

16. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Spring

17. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Freshwater Storage

18. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Plant uptake

19. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Groundwater Discharge

20. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Water storage in oceans

21. (1.00 pts) Label the boxes above, the number for the box corresponds to the question.

Groundwater Storage

Fill in the blank for questions 22-27

22. (1.00 pts) _____ occurs when water vapor cools and changes into liquid water droplets that form clouds in the atmosphere.

Condensation

23. (1.00 pts) _____ occurs when liquid water from the Earth's surface and from living organisms changes into water vapor.

Evaporation

24. (1.00 pts) _____ is the downward movement of water through pores and other spaces in the soil due to gravity.

Percolation

25. (1.00 pts) _____ is the movement of water into the ground due to the pull of gravity.

Infiltration

26. (1.00 pts) _____ is water that flows across land and collects in rivers, streams, and eventually the ocean.

Runoff

27. (1.00 pts) _____ is rain, snow, sleet, or hail that falls from clouds onto the Earth's surface.

Precipitation

28. (1.00 pts) What does USGS stand for + when was it founded?

Expected Answer: The United States Geologic Survey, founded in 1879

29. (2.00 pts) How did water play a role in the creation of Death Valley?

Expected Answer: The salt beds indicate the presence of water. Also it was a lake filled with water from the ice age. It turned into a desert probably towards the end of the ice age and also the fact that the rocks are made of limestone and sandstone indicate the presence of a shallow sea.

30. (3.00 pts) Compare hard and soft water. (*)

Expected Answer: Hard water consists of concentrations of Ca2+ and MG2+. Softwater involves the removal of hardness ions, sometimes with distillation. Hard Water has certain positive effects like getting calcium or magnesium. It is also bad because it can lead to the consumption of toxic metals or heavy metals.

Examine the diagram below, and answer questions 31-32

31. (1.00 pts) The wide curves in the river in the diagram are called

- (●) A) Meanders
- ( ) B) Watersheds
- ( ) C) Tributaries
- ( ) D) channels

32. (1.00 pts) The river in the diagram is most likely in the ____ stage.

- ( ) A) Youthful
- (●) B) Mature
- ( ) C) Old
- ( ) D) rejuvenated

33. (1.00 pts) As a river passes from a youthful to a mature stage, it

- ( ) A) Shortens
- ( ) B) Deepens
- ( ) C) Narrows
- (●) D) Meanders

34. (1.00 pts) At a meander of a river, erosion occurs

- (●) A) On the outside of the bank where the water flows faster.
- ( ) B) On the outside of the bank where the water flows slower.
- ( ) C) Along the inside bank where the water flows slower.
- ( ) D) Along the inside bank where the water flows faster.

35. (1.00 pts) At a meander of a river, deposition occurs

- A) On the outside bank where the water flows faster.
- B) On the outside bank where the water flows slower
- C) Along the inside bank where the water flows slower
- D) Along the inside bank where the water flows faster.

36. (1.00 pts) Alluvial fans and deltas have similar

- A) Locations and depositions
- B) Sizes of sediment grains
- C) Shapes of deposition
- D) Angles of slope

37. (1.00 pts) One difference between an alluvial fan and a delta is that an alluvial fan is

- A) Deposited on dry ground
- B) Deposited in water
- C) Relatively flat
- D) Formed by stream deposition

38. (1.00 pts) Channel erosion would most likely be quicker in streams with

- A) Bed Load
- B) Suspended Load
- C) Dissolved Load
- D) Mineral Load

39. (1.00 pts) ____ are smaller streams or rivers that flow into larger ones.

Tributaries

40. (1.00 pts) Drainage basins are separated from each by other by an area called a

- A) Load
- B) Divide
- C) Channel
- D) Watershed

41. (1.00 pts) A drainage basin is also called a ____.

Watershed

42. (1.00 pts) Which of the following is the largest drainage basin?

- A) Mississippi River Basin
- B) Colorado River Basin
- C) Amazon River Basin
- D) Rio Grande Basin

43. (1.00 pts) The path that a stream follows is called a _____.

Channel

44. (1.00 pts) Which river will cause the most erosion?

- A) One with low gradient and low discharge
- B) One with high gradient and low discharge
- C) One with low gradient and high discharge
- D) One with high gradient and high discharge

45. (1.00 pts) The ___ consists of dissolved materials, such as sodium and calcium.

Dissolved Load

46. (1.00 pts) What is the difference between the Strahler Stream Order method and the Shreve Order method?

Expected Answer: In the Shreve Stream Order, you simply add the numbers as you go along, whereas in the Strahler Stream Order you increment by 1 regardless of the order

Label the correct terms for the different layers representing groundwater zones:

47. (1.00 pts) Identify the first layer of the diagram.

Surface Layer

48. (1.00 pts) Identify the second layer of the diagram.

Zone of Aeration

49. (1.00 pts) Identify the third layer of the diagram.

Water Table

50. (1.00 pts) Identify the fourth layer of the diagram.

Zone of Saturation

51. (1.00 pts) Groundwater accounts for nearly ___ percent of The United States freshwater resources (*)

95

52. (1.00 pts) Porosity is dependent on all of the following except...

- A) Particle shape
- B) Sorting
- C) Amount of cement
- D) Hardness

53. (1.00 pts) What is the measure of how easily water can flow through an aquifer?

Permeability

54. (1.00 pts) A soil sample with a large amount of space between the particles will have a...

- A) High Porosity
- B) High Capillarity
- C) Low permeability rate
- D) Low infiltration rate

55. (1.00 pts) Flash flooding often occurs in city areas because:

Expected Answer: Roads, pavements, and buildings reduce the infiltration of water into the ground

56. (2.00 pts) If a rock is compressed and its volume decreases, how will the rock's density and porosity be affected?

Expected Answer: Density will increase and porosity will decrease

57. (1.00 pts)

The diagram represents three containers, A, B, and C, which were filled with equal volumes of uniformly sorted plastic beads. Water was poured into each container to determine porosity and infiltration time. See picture for question

- A) 4
- B) 3
- C) 2
- D) 1 ◉

58. (1.00 pts) Which sample probably has the greatest porosity? (The arrows represent possible infiltration of rainwater)

- A) Granite Bedrock
- B) Pebble Soil ◉
- C) Pebble-and-sand soil
- D) Conglomerate bedrock

Examine the illustration below and use it to answer questions 59-62

59. (1.00 pts) The rock layer I is called the _____.

Cap rock

60. (1.00 pts) The rock layer II is called the _____.

Aquifer

61. (1.00 pts) The formation labeled X is called a(n) _____.

Artesian Spring

62. (1.00 pts) The diagram above illustrates a(n) _____.

Artesian formation

Identifying Gradient Patterns

63. (2.00 pts) How are drainage basins formed?

Expected Answer: A drainage basin is formed by the action of water as it forms streams and rivers that flow downhill.

64. (1.00 pts) Identify the drainage pattern.

Dendritic Pattern

65. (2.00 pts) How is the previous pattern formed?

Expected Answer: Dendritic pattern develops on relatively uniform bedrock.

66. (1.00 pts) Identify the drainage pattern.

Parallel Drainage Pattern

67. (1.00 pts) Identify the drainage pattern.

Trellis Drainage Pattern

68. (2.00 pts) How is the previous pattern formed?

Expected Answer: Trellis Pattern develops in areas of alternating weak and resistant bedrock.

69. (1.00 pts) Identify the drainage pattern.

Rectangular Drainage Patte

70. (2.00 pts) How is the previous pattern formed?

Expected Answer: Rectangular pattern develops on highly jointed bedrock.

71. (1.00 pts) Identify the drainage pattern.

Radial Drainage Pattern

72. (2.00 pts) How was the previous pattern formed?

Expected Answer: Radial pattern develops on isolated volcanic cones or domes.

73. (1.00 pts) Identify the drainage pattern.

Centripetal Drainage Patter

74. (1.00 pts) Identify the drainage pattern.

Deranged/Contorted Draina

75. (1.00 pts) What type of stream pattern would most likely be found on the type of landscape shown in the diagram?

a. 

b. 

c. 

d. 

- (●) A)
- ( ) B)
- ( ) C)
- ( ) D)

76. (1.00 pts) The figure above shows a stream drainage pattern. Arrows show the direction of stream flow. In which landscape region did this drainage pattern develop?

a.

b.

c.

d.

- ( ) A)
- ( ) B)
- (●) C)
- ( ) D)

77. (1.00 pts)
The block diagram below represents the under-lying bedrock structure of a landscape region. Which diagram represents the most likely stream drainage pattern on the surface of this landscape?

a.

b.

c.

d.

- A)
- B)
- C)
- (●) D)

78. (1.00 pts) The diagram below shows a region that has undergone faulting. Which stream drainage pattern is the most likely to develop on the surface of the region?

a.

b.

c.

d.

- (●) A)
- B)
- C)
- D)

79. (2.00 pts) Will the sinuosity index of a river be higher or lower in regions of steep topography?

Why? (*)

Expected Answer: Lower, because rivers on slopes tend to cut straight channels.

80. (2.00 pts) What is the difference between stream competence and capacity?

Expected Answer: Competence is the largest material the stream can move, capacity is the maximum load of sediment it can carry.

81. (1.00 pts) How is a solution valley formed?

Expected Answer: It occurs when several sinkholes merge to form a larger depression.

82. (1.00 pts) The diagram below shows some features in a cave. Which type of rock was chemically weathered by acidic groundwater to produce the cave and its features?

- A) Siltstone
- B) Quartzite
- C) Basalt
- D) Limestone

83. (1.00 pts) Which graph best represents the relationship between the discharge of a stream and the velocity of stream flow?

a.

b.

c.

d.

- A)
- B)
- C)
- D)

84. (1.00 pts)
A stream flowing at a velocity of 250 cms per second is transporting sediment particles ranging in size from clay to cobbles. Which transported particles will be deposited by the stream if its velocity decreases to 100 cms per second?

- A) Cobbles, only

- (B) Cobbles and some pebbles, only
- (C) Cobbles, pebbles, and some sand, only
- (D) Cobbles, pebbles, sand, silt, and clay

85. (1.00 pts) The largest particles that a stream deposits as it enters a pond are 8 cm in diameter. The minimum velocity of the stream is approximately (in cm/s)

200 cm/sec

86. (1.00 pts) Which graph represents the general relationship between soil particle size and the permeability rate of infiltrating rainwater?

a.

b.

c.

d.

- (A)
- (B)
- (C)
- (D)

For questions 87-90 use the map below, which shows watershed regions of NY.

87. (1.00 pts) On which type of landscape region are both the Susquehanna-Chesapeake and the Delaware watersheds located?

Expected Answer: plateau

88. (1.00 pts) Most of the surface bedrock of the Ontario-St. Lawrence watershed was formed during which geologic time period?

- (A) Precambrian and Cambrian
- (B) Missisipian, Pennsylvanian, and Permian

C) Triassic, Jurassic, and Cretaceous

D) Ordovocian, Silurian, and Devonian

89. (1.00 pts) Outwash plains are formed as a result of deposition by

A) Ocean waves

B) Winds from hurricanes

C) Meltwater from glaciers

D) landslides

90. (1.00 pts) Which graph best represents the relationship between surface-water runoff and stream discharge?

a.

b.

c.

d.

A)

B)

C)

D)

Congrats, you made it! Thank you for completing the Rickards Dynamic Planet C Test!

© 2022 - powered by Scilympiad (https://scilympiad.com) | Tournaments (/rickards/Home/Subscribers?sort=t)
Terms of Use (/ToS.html) | Privacy (/Privacy.html) | Cookie Policy (/CookiePolicy.html) | Support (/rickards/Support) | Contact (/rickards/Home/Contact)